Mini Systematic Review: Session 4 Deliverable

Research Methods & Scientific Integrity in AI, Doctoral Program in Deep Tech, UNMSM

Mayté Ciriaco Ruiz

24 October 2026

Review question

This mini systematic review asks which LLM-based multi-agent architectures have been published in peer-reviewed journals (2020–2026) to extract, classify or verify information in unstructured text, and how they handle public-document contexts, Spanish or Latin American settings, traceability and ethical governance. It follows the design science paradigm of the author's thesis.

CIMO element	Scope in this review
Context	Unstructured text documents and claims; public, administrative, legal or civic documents; Spanish and Latin American civic and journalistic settings
Intervention	Multi-agent architectures built on large language models (two or more interacting agents with distinct roles)
Mechanism	Role decomposition, debate, evidence retrieval, evidence-linked claims, verification layers
Outcome	Quality of extraction, classification or verification; traceability; governance and accountability treatment

The thesis question that frames the review: under what design and evaluation conditions can a multi-agent system based on LLMs extract, classify and verify information in unstructured public documents in Spanish, while keeping traceability and ethical governance criteria applicable to civic and journalistic contexts in Latin America?

Search strategy

Scopus was searched on 24 October 2026 in three iterations; the third (44 records) is the final strategy. Spanish, Peru/Latin America and document type were not used as search filters, because iteration 1 returned zero records; they are applied as criteria and tie-break during screening.

Iteration	Blocks	Filters	Records
1	Multi-agent + extraction/verification + public documents + Spanish/Peru/Latin America	2020–2026; article	0
2	Multi-agent + extraction/verification	2020–2026; article	72

Iteration	Blocks	Filters	Records
3 (final)	Multi-agent + LLM + extraction/verification	2020–2026; article; journal sources	44

Iteration 3, full string:

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TITLE-ABS-KEY(("multi-agent" OR "multi-agent system*" OR "LLM agent*") AND ("large language model*" OR LLM OR LLMs) AND ("information extraction" OR "document verification" OR "fact-checking" OR "fact verification" OR "claim verification")) AND PUBYEAR > 2019 AND PUBYEAR < 2027 AND DOCTYPE(ar) AND SRCTYPE(j)
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Iteration 2, full string:

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TITLE-ABS-KEY(("multi-agent" OR "multi-agent system*" OR "LLM agent*") AND ("information extraction" OR "document verification" OR "fact-checking" OR "fact verification" OR "claim verification")) AND PUBYEAR > 2019 AND PUBYEAR < 2027 AND DOCTYPE(ar)
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Iteration 1, full string:

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TITLE-ABS-KEY(("multi-agent" OR "multi-agent system*" OR "LLM agent*") AND ("information extraction" OR "document verification" OR "fact-checking" OR "fact verification" OR "claim verification") AND ("public document*" OR "government document*" OR "administrative document*" OR "legal document*" OR "civic") AND (Spanish OR "low-resource language*" OR Peru OR "Latin America")) AND PUBYEAR > 2019 AND PUBYEAR < 2027 AND DOCTYPE(ar)
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Sources excluded by course rule: Google Scholar, preprints and grey literature.

Eligibility criteria (v1.0, dated 24 October 2026)

The criteria were written and saved before any include or exclude decision. The author had seen a list of titles on the Scopus results screen beforehand but made no decision at that point.

Code	Inclusion criterion (all must be met)
I1	Peer-reviewed journal article (Scopus DOCTYPE ar, SRCTYPE j), published 2020–2026, with a DOI
I2	Full text in English or Spanish and accessible to the reviewer
I3	Uses large language models as a core component
I4	Uses a multi-agent architecture: two or more interacting agents with distinct roles, not a single-model pipeline
I5	Addresses information extraction, classification or verification (fact, claim or document verification)
I6	Operates on unstructured text documents or claims, not on images, sensors or tabular data only

Priority tiers (applied after I1–I6): Tier 1 = public, government, administrative, legal or civic documents; Tier 2 = other unstructured text (for example news or scientific text) in any language.

Tie-break rule, in order: (1) tier; (2) Spanish, Latin American or low-resource-language context; (3) explicit treatment of traceability, provenance or ethical governance; (4) most recent year.

Code	Exclusion reason (one per record; first failing criterion in this order)
E1	Not a journal article or no DOI
E2	Language not English or Spanish
E3	Not LLM-based
E4	Not multi-agent
E5	No extraction, classification or verification task
E6	Out of domain (non-text data)
E7	Full text not retrievable
E8	Duplicate record
E9	Eligible but not selected or not assessed (see Limitations)

Screening procedure and declaration of AI use

An AI assistant (Claude, Anthropic, via claude.ai, 24 October 2026) was the primary screener at the title and abstract stage. It applied criteria v1.0 to the title, abstract and author keywords of all 44 records and proposed a decision and a reason code for each. Abstracts that did not settle a criterion were kept as Maybe and carried to full text. The prompt was the criteria in the previous section, with the reason-code precedence stated there.

The human reviewer validated a random sample of five records (drawn with Python random.Random, seed 20261024, from IDs 1–44) plus record 18, which the AI had flagged as low confidence. That is six of 44 records (13.6%), above the 10% minimum; the reviewer's Agree or Disagree for each is recorded in the screening log, sheet Validation. Disagreements are resolved in favour of the human reviewer. Cohen's kappa is not reported, because the AI acted as primary screener and not as an independent second reviewer.

Full-text eligibility, the tier, context and governance ratings, and the final selection of ten were done by the human reviewer. The screening log (Excel) with every decision, rationale and reason code is stored in the repository.

PRISMA 2020 flow numbers

From 44 records identified in Scopus, ten studies are included. The numbers below are the values to enter in the official PRISMA 2020 flow diagram app; each phase adds up.

Phase	Box	n
Identification	Records identified from databases (Scopus)	44
Identification	Records removed before screening (duplicates, automation tools,	0

Phase	Box	n
	other)	
Screening	Records screened	44
Screening	Records excluded	29
Screening	Reports sought for retrieval	15
Screening	Reports not retrieved (E7, publisher paywall: log IDs 5 and 34)	2
Eligibility	Reports assessed for eligibility	13
Eligibility	Reports excluded	3
Included	Studies included in review	10

Records excluded at screening (29): E1 no DOI, 1; E2 language, 1; E3 not LLM-based, 1; E4 not multi-agent, 5; E5 no extraction, classification or verification task, 4; E6 non-text data, 2; E9 eligible by abstract but not assessed at full text under the prioritised-reading rule, 15.

Reports excluded at full text (3): E2, full text in Chinese, 1 (log ID 33); E4, not multi-agent, 1 (log ID 19); E6, 1 (log ID 7, Gebreab et al., 2026: the abstract says the agents' claims are grounded in specific EHR fields and checked against the patient record, and the reviewer's full-text reading found that free-text notes are not processed, section 7.4.3).

Check: 44 − 29 = 15 sought; 15 − 2 = 13 assessed; 13 − 3 = 10 included.

Included studies (10)

Ten journal articles met criteria I1–I6 at full text and were selected by the tie-break rule. Scores follow the dated formula: 1000 for Tier 1, 100 for Spanish/Latin American/low-resource context, 10 for explicit traceability or governance, plus the publication year as a small final tie-break.

#	Study	Tier	Context	Governance	Documents processed
1	Iwaszenko, Czaja & Kozłowski (2026), *Applied Sciences*, doi:10.3390/app16094562	1	No	Yes (reviewer's reading of the full text)	Administrative and scientific document types of a national research institute; on-premise system for internal documents
2	Song, Dong, Dong & Liu (2026), *Journal of Business Research*, doi:10.1016/j.jbusres.2026.116242	1	No	Yes	ESG disclosures of 1,495 listed Chinese firms compared with negative news and regulatory records (5,031-report dataset); Tier 1 because of the regulatory records, while the reports are private disclosures
3	Nguyen, Vo & Tran (2026), *ICT Express*, doi:10.1016/j.icte.2026.05.	2	Yes	Yes	Claims in ViFactCheck, X-Fact and FEVER

#	Study	Tier	Context	Governance	Documents processed
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4	Paetzold et al. (2026), *Electronic Markets*, doi:10.1007/s12525-026-00928-6	2	No	Yes	Project emails, minutes, issue trackers
5	Dong & Ito (2026), *Scientific Reports*, doi:10.1038/s41598-026-41862-z	2	No	Yes	Textual claims
6	Zhang et al. (2026), *Agriculture*, doi:10.3390/agriculture16030356	2	No	Yes	Livestock health scientific articles and monographs from Chinese-language sources, plus 500 image-caption pairs
7	Elmitwalli et al. (2025), *Frontiers in Artificial Intelligence*, doi:10.3389/frai.2025.1659861	2	No	Yes	Tobacco and health claims checked against WHO, CDC, PubMed Central, Cochrane
8	Xiong et al. (2025), *Information Processing & Management*, doi:10.1016/j.ipm.2025.104241	2	No	Yes	Claims and evidence reports (RAWFC, LIAR-RAW)
9	Gorshenin & Fedorov (2025), *Pattern Recognition and Image Analysis*, doi:10.1134/S1054661825700543	2	No	Yes	Fake messages in text, images and audio (real and synthetic)
10	Pu et al. (2026), *Journal of Manufacturing Systems*, doi:10.1016/j.jmsy.2026.04.031	2	No	No	Aviation assembly process documents

Context = Spanish, Latin American or low-resource-language context. Governance = explicit treatment of traceability, provenance or ethical governance.

Evidence passages. Quotations marked (abstract) are taken from the Scopus abstract record. Passages with a page number come from the full text, checked against the PDF (records 30 and 9).

- Iwaszenko et al. (abstract): “Built upon a LangGraph architecture, the system utilizes open-weight large language models (LLMs) to perform zero-shot document classification, dynamic routing, and specialized information extraction.”
- Song et al. (abstract): “We present Agent-Greenwashing, a multi-agent LLM framework for comparing corporate ESG disclosures with negative news and regulatory records.”
- Nguyen et al. (abstract): “a training-free multi-agent debate framework that coordinates multiple LLMs through structured critique, context summarization, and judge-guided consensus reasoning.”
- Paetzold et al. (full text): the Extraction Layer is a multi-agent system with three agent roles, the orchestrator agent, the specialized risk agents (financial, technical and stakeholder agents in the prototype) and the aggregator agent, “each with a distinct configuration and set of tools” (p. 8; roles in Table 4, p. 11). The agents run as a directed graph with typed message passing (p. 12), on emails, meeting minutes and issue trackers (p. 8).
- Dong & Ito (abstract): “we propose a framework called multi-agent fact-checking (MAFC), which includes multiple agents with unique information sources to measure the text’s credibility.”
- Zhang et al. (full text): “Each subtask is handled by a dedicated heterogeneous agent, and the agents are organized into a traceable directed execution graph using LangGraph’s state-driven orchestration mechanism” (section 2.2.1, p. 8 of 31). The five agents (Coordination, Entity Extraction, Ontology Normalization, Relation Extraction, Knowledge Fusion) and their roles are described in Appendix A.1 (pp. 27–28). Traceability and human supervision of new labels are discussed in section 4.3 (pp. 25–26).
- Elmitwalli et al. (abstract): “We constructed a three-agent pipeline using OpenAI GPT-4.1 and the Crewai framework.” The agents are a Content Analyzer, a Scientific Fact Verifier and a Health Evidence Assessor.
- Xiong et al. (abstract): “DelphiAgent, an innovative agentic framework for trustworthy fact-checking that employs multiple LLMs to emulate the workflow of the Delphi method.” Its decision module uses multiple LLM-based agents with distinct personalities.
- Gorshenin & Fedorov (abstract): “The architecture includes an intelligent router, modality-specific agents, and a context-weighted aggregation module.” The abstract adds that the system works in zero-shot mode with open pretrained models including YandexGPT-5-Lite-8B-Instruct and Qwen3-8B (journal pages 741–753).
- Pu et al. (abstract): “a large language model (LLM)-based multi-agent collaborative triple information extraction method is proposed.”

Gap table

There is one candidate gap for each of the four types. Each describes a limit of the ten included studies, not an absence in the whole literature: one database, journal articles only, and 15 records not assessed at full text.

Gap type	Candidate gap (reviewer inference)	Evidence from included studies	Citation
Knowledge	It is not known how multi-agent extraction and verification perform on authentic, non-benchmark documents, or when their outputs need human review.	A tobacco-claims pipeline was validated on 20 claims; the debate and Delphi-style verifiers are evaluated on public fact-checking benchmarks (ViFactCheck, X-Fact, FEVER; RAWFC, LIAR-RAW).	Elmitwalli et al. (2025); Nguyen et al. (2026); Xiong et al. (2025)
Methodological	Evaluate multi-agent extraction on authentic administrative and legal corpora, not on synthetic or single-category data.	The risk-extraction architecture is evaluated on 150 synthetic, case-grounded documents (p. 15), and its authors state that generalizability to authentic project communication "remains to be demonstrated" (p. 3). The document-workflow system reports classification accuracy across administrative and scientific document types, and demonstrates its multi-stage processing on scientific article summarization (abstract).	Paetzold et al. (2026); Iwaszenko et al. (2026)
Contextual	Test transfer to public-access documents in Spanish from Latin American institutions, such as Peruvian public records.	The two Tier 1 studies use the internal documents of one research institute and records of Chinese listed companies and regulators, not Spanish public-access documents. The only study with a non-English evaluation context tests general claim benchmarks (Vietnamese, X-Fact).	Iwaszenko et al. (2026); Song et al. (2026); Nguyen et al. (2026)
Theoretical	There is no shared construct for what traceability and accountability mean in multi-agent document pipelines.	Each study implements it differently: versioned provenance with an extraction control flow (the authors call their four design principles nascent), Delphi-style consensus, credibility scoring, probabilistic aggregation with confidence calibration and interpretable justifications.	Paetzold et al. (2026); Xiong et al. (2025); Dong & Ito (2026); Gorshenin & Fedorov (2025)

Limitations and declared deviations

The main limit is that fifteen records were not assessed at full text because of the delivery deadline, so the ten included studies are the highest-priority eligible records, not the best of all eligible ones.

- Prioritised reading (declared deviation, adopted 24 October 2026): full texts were assessed in a fixed order (Tier 1 candidates, then tie-break contenders, then Tier 2 records) and reading stopped once ten eligible studies were confirmed. Records eligible by abstract but not reached are reported as E9, not assessed, and not as failing a criterion.
- Two reports were not retrieved because of publisher paywalls (log IDs 5 and 34, both strong candidates for a future update).
- Verification status: the passages for records 30 and 9 were checked against the full-text PDFs. For the other included records, the quotations come from the abstracts, and their full-text eligibility rests on the reviewer's reading and notes; a systematic re-check of the quotations against the PDFs is left for the next stage.
- One database (Scopus), English keywords, and abstract-level screening by an AI assistant with a 13.6% human validation sample.
- Tier 1 here does not mean public access: record 37 covers internal documents of one institute and record 43 compares private corporate disclosures with regulatory records.
- Google Scholar, preprints and grey literature were excluded by course rule.
- Gap statements describe the ten included studies and are reviewer inferences.